8 Measurement

Tune In

The correct statements are: I have two water bottles with a capacity of 1 L each.
- Oh! I am so tired as my bag weighs 3 kg.
- We are about 120 km far from our school.

Let's Try! 1

1. Option a. Length of your foot (will be measured in) cm.
2. Option b. Distance from your front door to the kitchen (will be measured in) metres.

Let's Try! 2

1. 45 cm = <u>450</u> mm
2. 70 mm = 7 cm
3. 999 mm = 99.9 cm

Let's Try! 3

1. 9 m = 9000 mm
2. 1200 mm = 1.2 m
3. 3.9 m = 390 cm
4. 784 cm = 7.84 m

Let's Try! 4

1. 3892 m **=** 3.892 km
2. 290 m = 0.290 km
3. 39 km = 39000 m
4. 8mm = 0.008 m

Exercise 8A

5.a. The length of my pencil is about 7 <u>cm.</u>
b. The height of a table about 2 <u>m</u>.

6.a. 3 cm < 45 mm < 4cm 6mm < 67 mm < 89 mm
b. 567 m < 2 km < 6 km 99 m < 7 km < 9 km 45 m
c. 123 m < 456 m < 890 m < 3 km < 6 km

7.
Robin : Length on the string = 6 cm
Rahman : Length on the string = 4 cm
Ram : Length on the string = 4.5 cm

a. Robin walked the longest distance.
Rahman walked the shortest distance.

b. Distance travelled by Robin = 6×50 m = 300 m
Distance travelled by Rahman = 4×50 m = 200 m
Distance travelled by Ram = 4.5×50 m = 225 m

Let's Try! 5

1.

cm	m
24	5
+ 15	8
40	3 (Ans)

2.

m	cm
9	05
− 5	18
3	87 (Ans)

Exercise 8B

1.a.

km	m
45	977
+ 6	367
52	**344**

b.

m	cm
36	35
+16	06
52	**41**

c.

km	m
3	675
2	150
+ 5	005
10	**830**

d.

cm	m
15	9
45	5
+ 8	6
70	**0**

2.a.

km	m
5	645
− 3	565
2	**080**

b.

m	cm
15	60
− 9	75
5	**85**

c.

cm	mm
98	9
− 22	8
76	**1**

d.

m	cm
497	12
− 289	06
208	**06**

Application Based Questions

3. Distance climbed by Manoj on first day of trek: 2 km 875 m
Distance climbed by him on second day of trek: 3km 295 m
Total distance climbed

km	m
2	875
+ 3	295
6	170

Ans: Manoj climbed a distance of 6 km 170 m on two days.

4. Distance cycled by Meena from home to school: 2 km 035 m
Distance cycled from school to music class: 780 m
Distance from music class to house: 1 km 415 m

Total distance cycled by Meena:

km	m
2	035
0	780
+ 1	415
4	230

Ans: Meena cycles a total of 4 km 230 m daily.

5. Measurement of Murali's jump: 8.31 m
Measurement of Montler's jump: 8.27 m

Difference in jump lengths: m cm

$$\begin{array}{r r} 8 & 31 \\ -\underline{8} & \underline{27} \\ \underline{0} & \underline{04} \end{array}$$

Ans: Murali Shreeshankar won the long jump by a margin of 4 cm.

6. Total distance between Delhi and Jaipur is the sum of the readings on the two sides of the
milestone.

$$\begin{array}{r r} \text{km} & \text{m} \\ 182 & 500 \\ +\underline{229} & \underline{300} \\ \underline{411} & \underline{800} \end{array}$$

Ans: The total distance between Delhi and Jaipur as per the milestone is 411 km 800 m.

Let's Try! 6

1. A sack of rice will be measured in **kg**.
2. A cup of sugar is measured in **g**.

Let's Try! 7

1. 2.389 kg = 2.389 × 1000 = 2389 g.(Ans)
2. 1750 g = 1750 / 1000 = 1.750 kg. (Ans)

Let's Try! 8

1. 3892 mg **=** 3892/1000 = 3.892 g
2. 52.91 g = 52.91 × 1000 = 52910 mg

Let's Try! 9

1.

$$\begin{array}{r r} \text{g} & \text{mg} \\ 78 & 841 \\ +\underline{52} & \underline{621} \\ \underline{131} & \underline{462} \end{array}$$ **(Ans)**

2.

$$\begin{array}{r r} \text{kg} & \text{g} \\ 66 & 962 \\ -\underline{31} & \underline{562} \\ \underline{35} & \underline{400} \end{array}$$ **(Ans)**

Exercise 8C

4.a.

kg	g
7	7000
12	12000
22	22000
3.9	3900

b.

g	mg
18	18000
2	2000
4	4000
0.5	500

5.a. 5 kg 750 g
= 5 × 1000 + 750 g
= 5750 g (Ans)

b. 4 kg 10 g
= 4 × 1000 + 10 g
= 4010 g (Ans)

c. 8 kg 355 g
= 8 ×1000 + 355 g
= 8355 g (Ans)

d. 10 kg 110 g
= 10 × 1000 + 110 g
= 10110 g (Ans)

e. 9 kg 950 g
= 9 × 1000 + 950g
= 9950 g (Ans)

f. 6 kg 50 g
= 6 × 1000 + 50 g
= 6050 g

6.a. 5000 mg
= 5000/1000 g
= 5 g (Ans)

b. 83000 mg
= 83000/1000 g
= 83 g (Ans)

c. 71000 mg
= 71000/1000 g
= 71 g (Ans)

d. 34860 mg
= 34860/1000 g
= 34.860 g (Ans)

e. 65 mg
= 65/1000 g
= 0.065 g (Ans)

f. 650 mg
= 650/1000 g
= 0.650 g (Ans)

7.a.
kg g
1 2 3 5 0
+ 4 0 6 5
1 6 4 1 5 (Ans)

b.
g mg
5 6 4 5 0
1 2 9 8 0
+ 7 2 2 5
7 6 6 5 5 (Ans)

c.
kg g
6 5 4 8 0
- 3 9 6 8 5
2 5 7 9 5 (Ans)

d.
g mg
8 6 0 4 7
- 5 9 4 6 0
2 6 5 8 7 (Ans)

e.
kg g
2 5 4 6 0
7 5 0 1 0
+ 4 0 7 5 0
1 4 1 2 2 0 (Ans)

f.
g mg
5 4 1 5 0
- 4 8 9 0 0
5 2 5 0 (Ans)

Application Based Questions

8. Weight of apples bought: 20 kg 500 g
Weight of pineapples : 18 kg
Weight of oranges : 22 kg 750 g
Weight of grapes : + 10 kg 250 g
Total weight of fruits 71 kg 500 g

Ans: The total weight of fruits is 71 kg 500 g

9. Weight of text books : 675 g
Weight of notebooks: 415 g
weight of pencil box: 125 g
Weight of lunch box: 205 g
Weight of file : 115 g

Weight of empty bag: 315 g
Total wt. of bag = 1850 g

Ans: The total weight of the bag with all its contents is 1850 g or 1 kg 850 g

10. Stock of rice at the beginning of the month : 25 kg 000 g
Stock of the rice at month end : 8 kg 600 g

Rice consumed in the month is 25.000 – 8.600 kg

$$\begin{array}{r} 25.000 \text{ kg} \\ -\ \underline{8.600 \text{ kg}} \\ \underline{16.400 \text{ kg}} \end{array}$$

Stock of Dal at the beginning of the month: 4 kg 500 g
Stock of Dal at month end: 750 g

Dal consumed during the month = 4. 500 kg

$$\begin{array}{r} \underline{-\ 0.750 \text{ kg}} \\ \underline{3.750 \text{ kg}} \end{array}$$

Ans: The quantity of rice consumed during the month was 16 kg 400 g and dal was 3 kg 750 g.

11. Weight of Karthik's sports kit is the total weight of its various items.

	kg	g
Weight of bats:	2	4 5 0
Weight of balls:	0	1 5 0
Weight of gloves:	0	0 8 0
Weight of shoes:	0	7 7 0
Weight of pads:	0	4 1 5
Total	3	8 6 5

Ans: Karthik's sports kit weighs 3 kg 865 g

12. Nimmi's sister's weight after two months: 4 kg 175 g
Nimmi's sister's weight at birth: 3 kg 250 g

Weight gained in two months: 4 . 1 7 5 kg

$$\begin{array}{r} \underline{-\ 3.250 \text{ kg}} \\ \underline{0.925 \text{ kg}} \end{array}$$

Ans: Nimmi's sister gained 0.925 kg in two months after birth.

Let's Try! 10

1. The volume of water in a cup is measured in millilitre.
2. The volume of water in a drum is measured in Litres.

Let's Try! 11

1.a. 3231 ml
= 3231/1000 L
= 3.231 L **(Ans)**

b. 42 mL
= 42/1000 L
= 0.042 L **(Ans)**

2.a. 5.3 L
= 5.3 × 1000 mL
= 5300 mL **(Ans)**

b. 7 L
= 7 × 1000 mL
= 7000 mL **(Ans)**

Exercise 8D

4.

L	6	8	3	4	23	64
mL	6000	8000	3000	4000	23000	64000

5.a. **L mL**
32 150
+ 8 760
40 910 (Ans)

b. L mL
45 900
+ 6 550
52 450 (Ans)

c. L mL
58 430
- 39 280
19 150 (Ans)

d. L mL
97 550
- 75 985
21 565 (Ans)

e. L mL
7 668
14 349
+ 23 563
45 580 (Ans)

f. L mL
39 075
- 28 335
10 740 (Ans)

Application Based Questions

6. Oil consumed by Maya: 2 L 560 mL = 2.560 L
oil consumed by Meera: 2650 mL or 2 L 650 mL = 2.650 L
Oil consumed by Mini: 2 L 605 mL = 2.605 mL
2.560 < 2.605 < 2.650
Ans: Thus, Meera uses the greatest amount of oil while Maya uses the least amount.

7.a. Petrol filled by Harish in March: 12 L 675 mL
Petrol filled by Harish in April: 5 L 350 mL
Total quantity of petrol filled: 18 L 025 mL
Ans: Amount of petrol filled in two months is 18 L 25 mL.

b. Advantages of car pool are – i) It reduces air pollution. ii) It saves fossil fuels which are non-renewable resources iii) It brings down the expenditure on travelling and on fuels.

8. Total amount of milk purchased: 2 L 500 mL = 2.500 L
Amount used for coffee: 750 mL = 0.750 L
Amount fed to cat: 350 mL = 0. 350 L

Amount of milk left over: 2.500 L – (0.750 + 0.350) L
= 2.500 – 1.100 L
= 1.400 L

Ans: Mrs Jacob is left with 1 L 400 mL of milk after having coffee and feeding the cat.

9. Amount of diesel in the barrel: 45 L
Amount poured in first can: 4 L 500 mL = 4.500 L
Amount poured in second can: 10 L 650 mL = 10.650 L
Amount poured in 3^{rd} can: 12 L 750 mL = 12.750 L

Amount left in the barrel: 45 L – (4.500 + 10.650 + 12.750) L
45 L – 4.500
10.650
+ 12.750
27.900 L

Amount left in barrel: 45.000
- 27.900
17.100 L

Ans: Amount of diesel left in barrel is 17 L 100 mL

10. a. Amount of milk used each day: 500 L 560 mL
Amount of cold water used: 800 L 780 mL
More amount of water is used:

Excess quantity of water over milk : 800 L 780 mL
- 500 L 560 mL
300 L 220 mL

c. Total quantity of rose syrup and milk = 10 L 250 mL + 500 L 560 mL
= 500 L 560 mL

11. Dosage of cough syrup Pratap has in a day: 5 mL + 5 mL + 5 mL = 15 mL
Amount of cough syrup consumed in 5 days: 5 x 15 mL = 75 mL

Quantity of syrup in the beginning: 200 mL
Quantity of syrup left after 5 days: 200 mL – 75 mL
= 125 mL

Ans: Pratap had 75 mL of syrup for 5 days and 125 mL was left in the bottle.

Revision Exercises

1.a. 2 km 130 m = 2130 m.

b. 534 cm = 5 m 34 cm.

c. 5045m = 5 km 45 m.

d. 56 cm = 5 cm 60 mm.

e. 5 kg 132 g = 5132 g.

f. 34521 mL = 34 L 521 mL

g. 3201 g = 3 kg 201 g

h. 4 L = 4000 mL.

2.a. 87 m = 87 × 100 cm = 8700 cm

b. 6m 15 cm = 6 × 100 +15 cm = 615 cm

c. 234 mm = 234/10 = 23 cm 4 mm or 23.4 cm

d. 88 km = 88 × 1000 = 88000 m

e. 600 cm = 6/100 = 6 m

f. 12 km 35 m = 12×1000 + 35 m = 12035m

g. 15 kg = 15 × 1000 = 15000 g

h. 40 kg 440 g = 40 × 100 + 440 = 40440 g

i. 51050 mg = 51050/1000 = 51.050 g

j. 11050 g = 11050/1000 = 11. 050 kg or 11 kg 50 g

k. 65g 10 mg = 65 × 1000 + 10 mg = 65010 mg

l. 65 mL = 65/1000 = 0.065 L

3.a.

  9 km 200 m
+ 3 km 500 m
 12 km 700 m **(Ans)**

b.

 9 m 15 cm
 6 m 70 cm
+ 6 m 25 cm
 22 m 10 cm **(Ans)**

c.

 6 kg 870 g
– 2 kg 745 g
 4 kg 125 g **(Ans)**

d.

 25 L 150 mL
- 12 L 578 mL
 12 L 572 mL **(Ans)**

Level Up

1.a. 89 cm, 4m 6 cm, 8m 8 cm, 109 cm, 203 cm

89 cm < 109 cm < 203 cm <4 m 6 cm < 8m 8 cm

b. 12 g, 100 mg, 200 mg, 671 mg, 10 g

100 mg < 200 mg < 671 mg < 10 g < 12 g

2.a. Distance covered by Sharma on first day: 3 km 550 m

Distance covered by Sharma on 2nd day: 5 km

Distance covered on third day: 2 km 675 m

Total distance covered on the 3 days: 11 km 225 m

Ans: Gopal Sharma walked a total of 11 km 225 m on the first three days.

b. Distance to be covered in half marathon: 21 km
Distance covered by Avinash: 16 km 245 m
Distance left to be covered: 21 – 16.245 km

$$\begin{array}{r} 21.000 \\ -\underline{16.245} \\ \underline{4.755} \text{ km} \end{array}$$

Ans: Avinash still has 4 km 755 m left to cover.

c. The total weight of the four pups is

$$\begin{array}{r} 145 \text{ g} \\ 154 \text{ g} \\ 165 \text{ g} \\ +\underline{180 \text{ g}} \\ \underline{644 \text{ g}} \end{array}$$

Weight difference between the heaviest and the lightest pups is:

$$\begin{array}{r} 180 \text{ g} \\ -\underline{145 \text{ g}} \\ \underline{35 \text{ g}} \end{array}$$

Ans: The total weight of pups is 644 g and the difference between the heaviest and lightest pups is 35 g.

d. Total capacity of the swimming pool: 62500 L
Water pumped before breakdown: 46535 L
Volume of water required to fill pool:

$$\begin{array}{r} 62500 \\ -\underline{46535} \\ \underline{15965 \text{ L}} \end{array}$$

Ans: 15965 L of water is required to fill the swimming pool to capacity.

e. Distance from A to B: 15 m 75 cm
Distance from B to C: 33 m 60 cm
Distance from C to D: 8 m 35 cm
Total distance A to D: 57 m 70 cm

Ans: The total distance travelled from Point A to D is 57 m 70 cm.

HOTS Questions

1.

J1	**J4**	**J5**	**J6**			
J2	**J3**		**J7**			
			J8	**J9**	**J10**	
J16	**J15**	**J14**	**J13**	**J12**	**J11**	
J17						
J18	**J19**	**J20**	**J21**	**J22**	**J25**	**J26**
				J23	**J24**	**J27**

In the given puzzle distance between each J is 25 cm.
As we trace the route from J1 to J27 the distance is
26 times 25 cm.

$$\begin{array}{r} 26 \\ \times\ 25 \\ \hline 130 \\ +\ 52\times \\ \hline 650 \text{ cm} \end{array}$$

Ans: The distance traveled from the first J to the last J in the puzzle is 6 m 50 cm.

2. Amount of water for each athlete: 200 mL
Amount of water for 40 athletes: 40 × 200 mL= 8000 mL
Converting into L: 8000 mL = 8000/1000 = 8 L
For 8 L, the school needed 8/2 = 4 bottles.
For one athlete, weight of 3 teaspoons of glucose: 3 × 4 g = 12 g
For 40 athletes, the amount of glucose used: 40 × 12 g = 480 g

Ans: For making energy drink of 40 athletes, 4 2L bottles of mineral water and 480 g of glucose was used.

Cross-curricular Connect

Tanker's total capacity: 7000 L
Water to first house : 3000 L
Water to second house: 2750 L
Water left in the tanker: 7000 L (3000 + 2750) L = 7000 L – 5750 L = 1250 L

Ans: After distributing water to two houses, 1250 L of water was left in the tanker.

Everyday Maths

1. Water spent in shower bath: 20 L 325 mL
Water used in bucket bath: 10 L

Water saved by Amayra: 20 L 325 mL

$$\begin{array}{r} -\ \underline{10 \text{ L } 000 \text{ mL}} \\ \underline{10 \text{ L } 325 \text{ mL}} \end{array}$$ **(Ans)**

2. Mass of Peter's travel bag: Mass of David's bag + excess weight
: 14 kg + 4 kg 250 g
= 18 kg 250 g.

Length of Peter's bag: Length of David's bag + excess length
= 50 cm + 150 mm
= 50 cm + 150/10 cm
= 50 + 15 cm
= 65 cm

Ans: The mass of Peter's bag is 18 kg 250 g and its length is 65 cm.